

2019-Ph-H1-Q16

Section: Electricity

Topic: Current, PD, Power, Resistance

Question:

Which row shows the correct combination of charge carriers and direction of current in a metal wire?

Charge carriers	Current direction
A. Electrons	Opposite to electron flow
B. Electrons	Same as electron flow
C. Protons	Same as proton flow
D. Ions	Opposite to ion flow
E. Protons	Opposite to proton flow

Answer:

In metals, the charge carriers are **electrons**, which move in the **opposite direction** to conventional current.

Correct Option: A (Electrons; opposite to electron flow)

Guidance for Students:

Remember, conventional current is from positive to negative — opposite to electron flow.

Revision Tips:

- Metals conduct via electrons, not protons
- Current direction = direction **positive charges would move**
- Don't confuse with ion flow in electrolytes